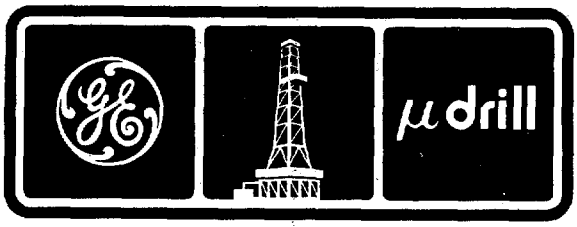
GEK-73911

DRIVE SYSTEM INSTRUCTIONS

μdrill™ 1500
RIG OPERATION MANUAL



DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

μdrill™ 1500
RIG OPERATION MANUAL

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The information contained herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, U.S.A.

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CHAPTER 1
INTRODUCTION

This manual describes the location, purpose, and operation of the μdrill™ 1500 drive system controls and indicators. A typical rig drive system includes a driller's console, power control equipment (often mounted in a power control room), a mud pump console (optional), and the necessary motors and generators. Figure 1 is a simplified diagram of a typical μdrill™ 1500 rig drive system.

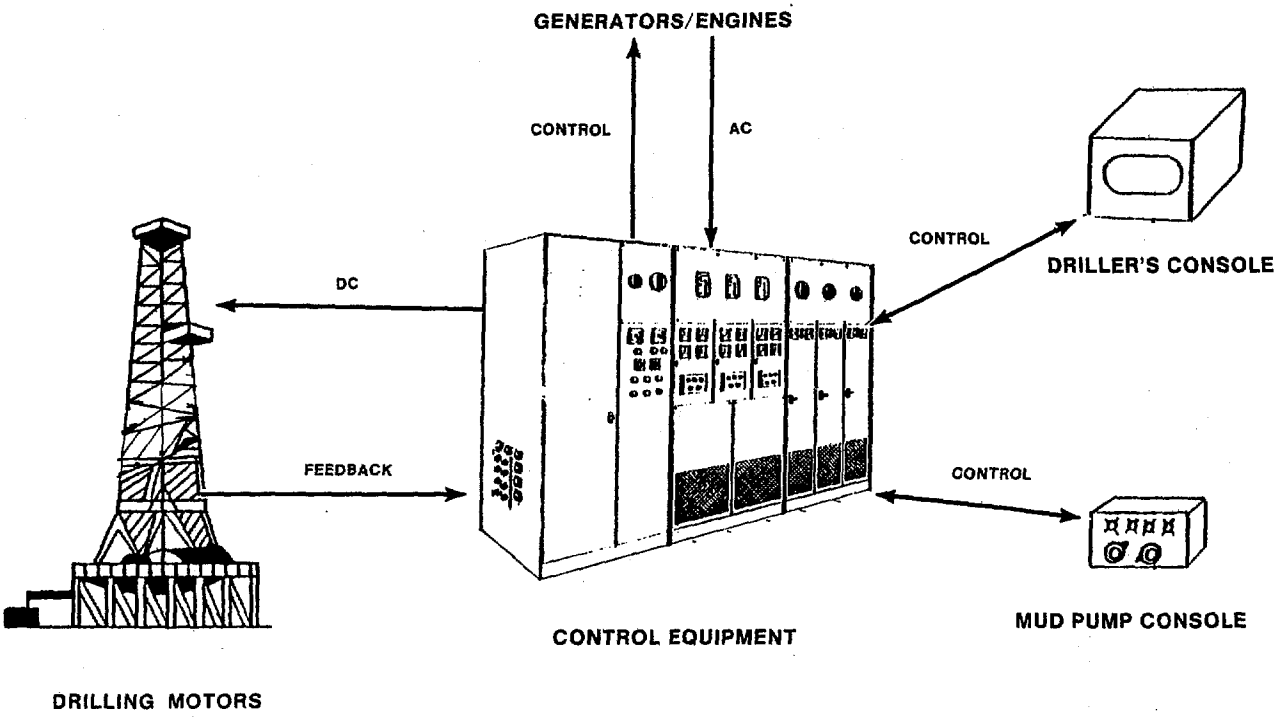


Figure 1.
Typical Rig Drive System

The **driller's console** contains controls such as assignment switches (used to connect specific motors to specific SCR power converters) and motor throttles. Indicators on the driller's console typically show equipment status such as motor load, SCR status, generator status, plus the status of oiler and blower motors.

The **mud pump console**, when used, contains a throttle and status lights for each mud pump. These controls duplicate those found on the driller's console. A switch on the driller's console determines whether the driller's console or mud pump console has control of the mud pump motors.

The **μdrill™ 1500 control equipment** contains a variety of controls and indicators, including generator and SCR controls, feeder breakers, ground detectors, and synchronizing equipment. These controls are mainly used for starting up and shutting down the rig drive system.

The μdrill™ 1500 rig drive system is available in a variety of configurations. There will be differences between rigs in such things as the number of motors, generators, and SCRs, the control functions available, whether the power control equipment is mounted in a power control room, and the types of options installed. Because of this, your rig drive system may not look exactly like some of the photographs or diagrams in this book. In addition, there may be descriptions of equipment or controls that your system does not have. Keep this in mind while using this manual, since all of the information may not apply to your particular rig.

CHAPTER 2
RIG CONTROLS AND INDICATORS

On the next several pages are photographs showing controls and indicators for the driller's console, mud pump console, and μdrill™ 1500 control equipment. Controls pointed out in the photograph are described in nearby text. See Chapter 3 for rig operating procedures.

DRILLER'S CONSOLE

The indicators, meters, throttles and switches on the driller's console give the driller complete control over the drilling motors. A description of each item on the driller's console is given below.

STATUS LIGHTS

These lights show the operating condition of rig equipment such as generators, SCRs, and motors. Each light is labelled with the name of the equipment or function monitored. When the light is on, the equipment or function is active. Some typical items that may be monitored by the status lights are listed below.

Gen On - The indicated generator is supplying power to the drive system when this light is on.

SCR On - The indicated SCR power converter is operating and can have a motor assigned to it when this light is on.

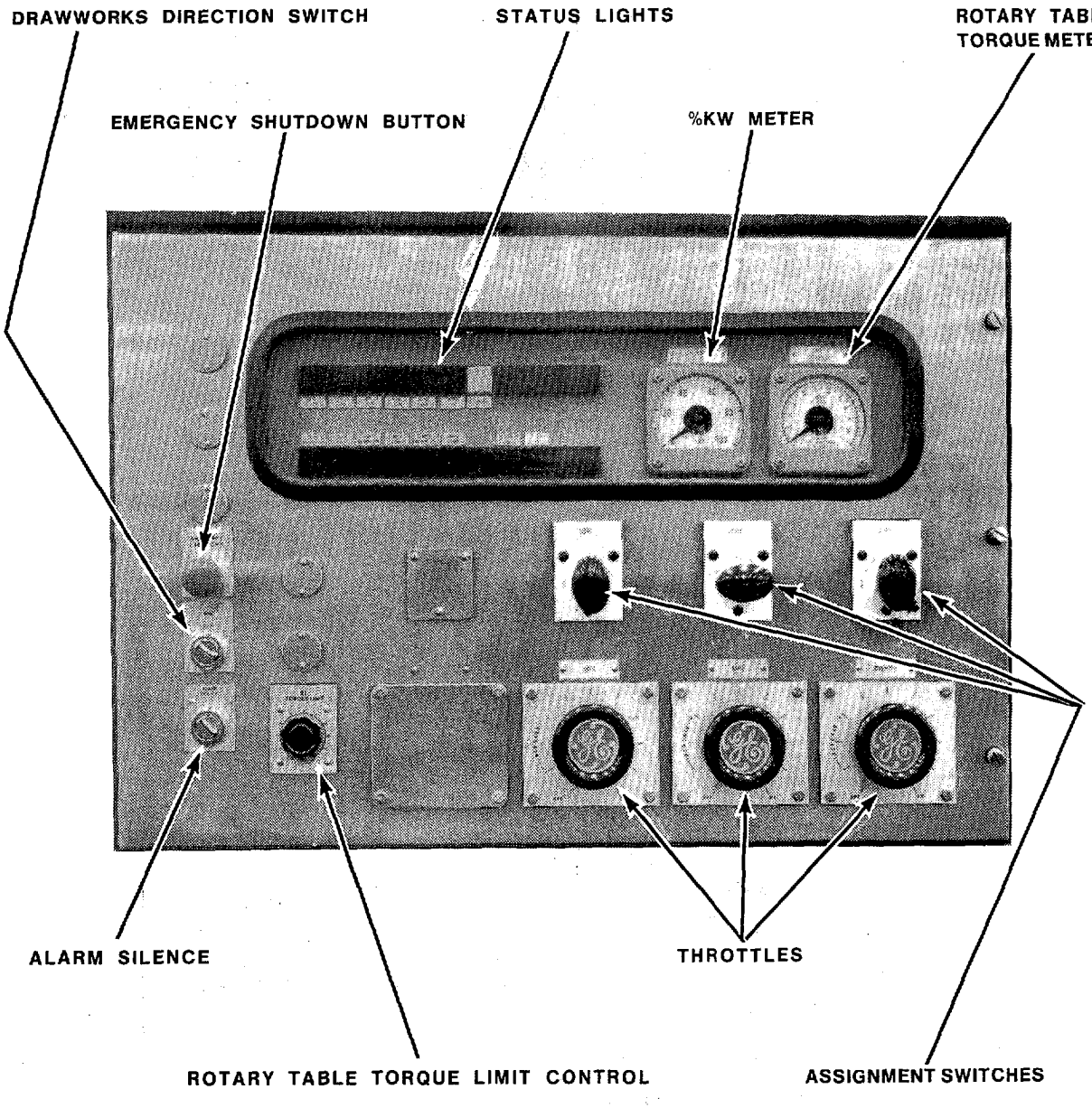
Blwr On - The indicated motor blower is operating and the assigned drilling motor is ready to run when this light is on.

Auxiliaries - The indicated lube motor for mud pump rod and chain oilers is operating when this light is on. (Optional)

Low Fuel - Fuel in the day tank at the engine is at a low level when this light is on. This condition could be caused by a clogged fuel line or a closed valve at the main fuel tank. (Optional)

High Load - The generators or diesels supplying power to the rig are approaching an overload condition when this light comes on.

Alarm Silenced - The Alarm Silence switch is on when this light is flashing. In this condition, the audible alarm will not sound for abnormal conditions.



(ICF51825)

Figure 2.
Typical Driller's Console

%KW METER

This meter indicates how much of the total power available from the engines is being used by the rig. Using this meter, an operator can determine the correct number of engines to run to supply enough power for rig operation. The position of the needle indicates whether one, two, or three engines are required.

ROTARY TABLE TORQUE METER

This meter indicates the amount of torque applied to the drill pipe by the rotary table. The meter may be calibrated both in foot-pounds and amperes.

ASSIGNMENT SWITCHES

These switches allow you to assign (connect) a specific drilling motor to a specific SCR power converter. The number of assignment switches depends on the number of power converters in the drive system; there will be one assignment switch for each power converter. The drilling motors that can be connected to each power converter are indicated on the switch nameplate. Automatic cross-interlocking prevents connecting the same motor to two SCRs.

THROTTLES

The throttles control the speed of the motors indicated on the respective console nameplates. The number of throttles depends on the number of drilling functions in your particular configuration. A typical configuration would include a throttle for each of two mud pumps and a third throttle for the drawworks and rotary table.

ROTARY TABLE TORQUE LIMIT CONTROL

The rotary table is driven by the drawworks motors. This control limits the total torque provided to the rotary table by the drawworks motors.

ALARM SILENCE

When activated, this switch prevents the audible alarm from sounding during abnormal conditions. The **alarm silenced light** will flash as long as this switch is ON.

DRAWWORKS DIRECTION

This switch determines the direction of rotation of the drawworks motors and is effective only when the drawworks throttle is OFF. The FWD and REV positions of the **drawworks direction switch** cause the drawworks motors to operate in the forward and reverse direction respectively. The FWD REG position enables regenerative braking of the drawworks motors. Regenerative braking is a method of slowing down a motor by pumping its energy back into the AC power system. This allows the motor to decelerate more rapidly. If selected, regenerative braking starts about one second after the drawworks foot throttle is released and the hoist drum is declutched from the drawworks.

MUD PUMP CONTROL HERE - REMOTE

These switches select whether the driller's console or mud pump console controls the mud pumps. There may be an indicating light on the mud pump console to indicate that the driller has relinquished control of the mud pumps. (Optional)

EMERGENCY SHUTDOWN

Depressing this button removes DC power from all drilling motors by deassigning them from any SCRs.

PLANT SHUTDOWN

Depressing this button shuts down all generator engines; the button must be pulled out before the engines can be restarted. The engines are normally shut down by the speed governor, but other methods such as manifold cut-out may be used. The button is mounted under a spring-loaded plastic cover to prevent accidental depression. (Optional)

AUDIBLE ALARM

Unless disabled by the **alarm silence switch**, the **audible alarm** sounds for several conditions that could cause a drill motor to stop. Some of these conditions could be:

1. Loss of a drill motor blower while the motor is assigned to an SCR (alarm only).
2. Loss of an SCR while a drill motor is assigned to it (will stop motor).
3. Loss of a mud pump auxiliary motor (i.e., a lube motor) while the mud pump motor is assigned to an SCR (alarm only).

MUD PUMP CONSOLE

The mud pump console is simply an extension of the driller's console and is used to control the mud pumps from a remote location. Figure 3 is a photograph of a typical mud pump console

THROTTLES

The mud pump throttles serve the same purpose on the mud pump console as on the driller's console: controlling the speed of the mud pump motors.

INDICATORS

There are two sets of indicators on the mud pump console, one set for each mud pump on the rig. These indicators are:

MP Control Here - When the **control here - remote switch** on the driller's console is in the REMOTE position, this light will come on, indicating that the mud pump console has control of the mud pump motors. There is one such indicator for each mud pump. (Optional)

MP Aux - When this light is on, the mud pump auxiliary (lube) motors are operating. There is one such indicator for each mud pump. (Optional)

MP Blower - Not used on this rig.

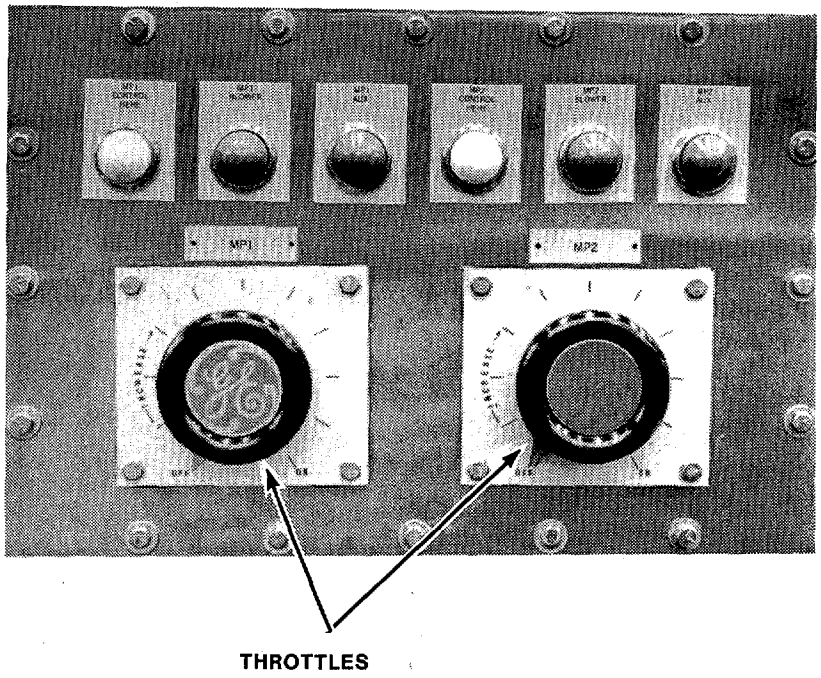


Figure 3.
Typical Mud Pump Console

CONTROL EQUIPMENT

The μdrill™ 1500 control equipment consists of one or more line-ups, often mounted in a power control room. Figure 4 is a diagram of a typical control equipment line-up with major controls pointed out. This equipment controls the engine/generator, converts three-phase AC power to DC power for the drilling motors, synchronizes the power supplied by several generators, and detects ground fault conditions in the AC or DC power circuits.

Most of the controls on the line-up are used during rig start up and shut down. The following panels and modules contain controls and indicators that the operator should be familiar with. Each of these panels and modules is illustrated and described in this manual —

- Engine/Generator Control Panel
- Synchronizing Panel
- Ground Detector Module
- Feeders
- Battery Module
- Motor Control Center
- Power Conversion Panel

ENGINE/GENERATOR CONTROL PANEL

The Engine/Generator Control Panel consists of indicators, meters, and switches used to start and control an emgine/generator. There is one Engine/Generator Control Panel for each generator set supplying power to the rig. Figures 5 and 6 are photographs of a typical control panel showing the following controls and indicators:

Voltmeter - Indicates the AC phase voltage between any two phases selected by the **voltmeter switch**. The voltmeter can measure AC bus voltage or phase to .

Ammeter - Indicates the current of the phase selected by the **ammeter** switch.

Kilowatt Meter - Indicates engine loading.

Kilovar Meter - Indicates generator loading. Reactive power is a measure of generator loading. High reactive power indicates a low power factor.

Engine Control Switch - Sets the operating mode of the generator engine. In the OFF position, the engine is shut down. In the IDLE position, the engine can be started and will operate at idle speed. In the RUN position, the engines operates at normal speed. In the LOAD position, the generator will load. The load sharing circuits of the generator are then connected with the load sharing circuits of other on-line generators so that all generators will equally share the electrical load.

Voltmeter Switch - Selects the voltage source to be displayed by the **voltmeter**. Selecting VL1-2 causes the voltmeter to display the voltage between phases 1 and 2 of the generator output. Likewise, selecting VL2-3 causes the voltmeter to display the voltage between phases 2 and 3 and selecting VL1 causes the voltmeter to display the voltage between phases 3 and 1. Selecting VBUS causes the VOLTMETER to display the voltage of the AC bus (phases 3-1).

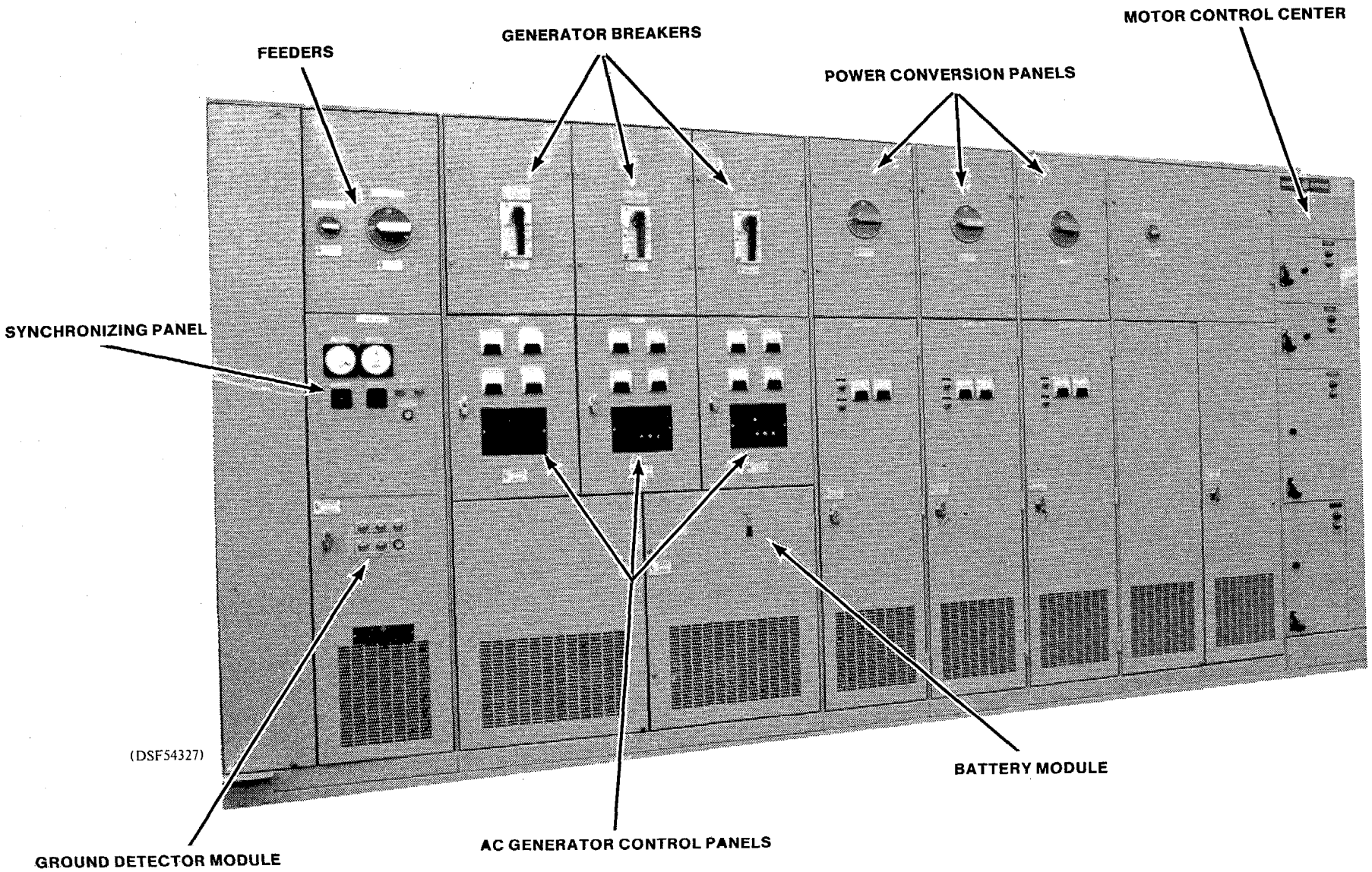


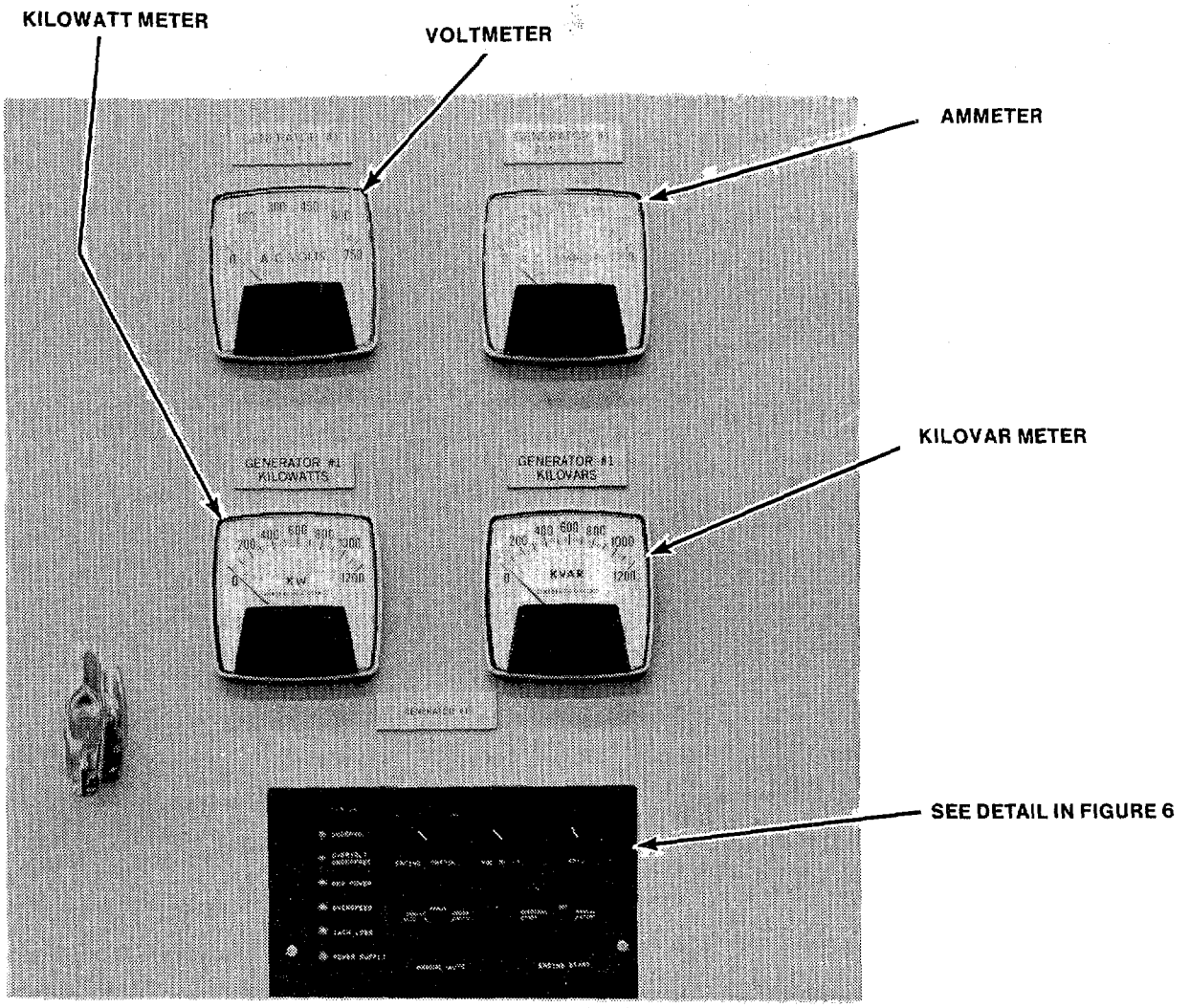
Figure 4.
Typical Control Equipment Line-Up

Ammeter Switch - Selects the phase current displayed by the **ammeter.**

Manual-Auto Switch - Not used on this rig.

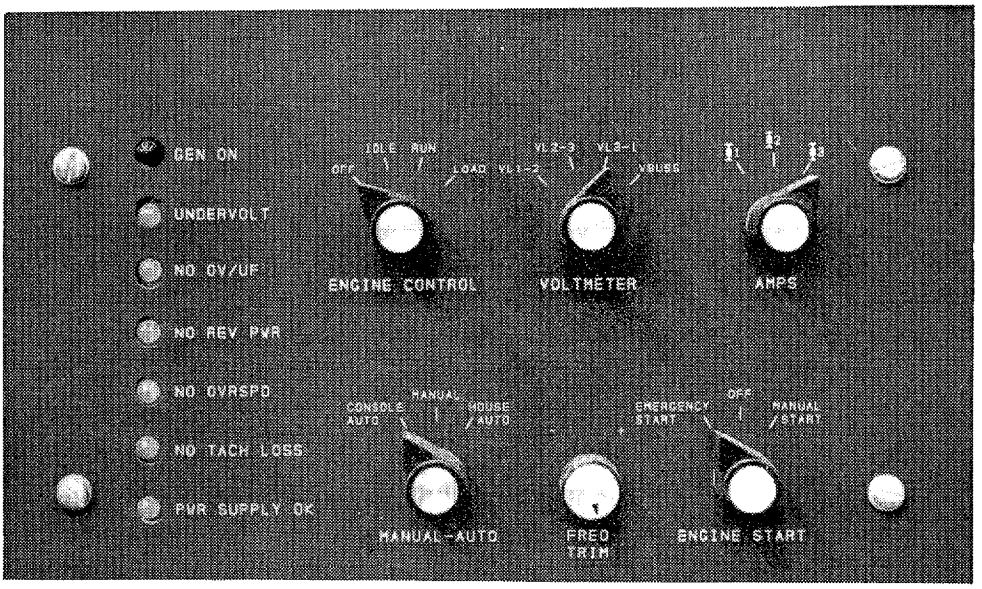
Frequency Trim Knob - Adjusts frequency of generator output.

Engine Start Switch - Not used on this rig.



(DSF53906)

Figure 5.
Engine/Generator Control Panel



(DSF54329)

Figure 6.
Engine/Generator Control Panel (Detail)

(NOTE: The following engine/generator control lights indicate whether the generator is operating normally or whether a fault has occurred. The first fault will cause its associated light to flash . Any subsequent fault that occurs will cause its associated light to go out.)

Gen On Light - When ON, indicates that the generator is running, the breaker is closed, and power is being supplied to the AC bus. When OFF, indicates that breaker is open.

Undervolt Light - When ON, indicates a generator voltage greater than 450V (on 600V generator). When OFF, indicates that a generator undervoltage condition exists. This condition causes the **generator circuit breaker** to trip.

No OV/UF Light - When ON, indicates a normal condition. When OFF, indicates that a generator overvoltage or underfrequency condition exists. The **generator circuit breaker** will trip for an overvoltage condition, but not for an underfrequency condition.

No Rev Pwr Light - When ON, indicates a normal condition. When OFF, indicates that reverse power was flowing from the AC bus back to the generator. This condition causes the generator circuit breaker to trip.

No Ovrspd Light - When ON, indicates a normal condition. When OFF, indicates that the engine/generator set was overspeeding. This condition causes the control to call for minimum fuel and trips breaker. NOTE: Mechanical overspeed should be provided with engines.

No Tach Loss Light - When ON, indicates a normal condition. When OFF. indicates that the tachometer output from the generator engine has been lost. This condition causes the engine/generator to shut down and trips breaker.

Pwr Supply OK Light - When ON, indicates a normal condition. When OFF, indicates that the generator control power supply has malfunctioned. This condition causes the generator engine to shut down and trips breaker.

SYNCHRONIZING PANEL

The controls and indicators on the synchronizing panel are used to synchronize the output frequency and phase of generators being brought on line with the frequency and phase of the 600 VAC bus. Failure to properly synchronize generators as they are brought on line can cause damage to the equipment. Figure 7 is a photograph of the synchronizing panel controls and indicators.

Frequency Switch - Selects the generator whose frequency is measured by the **frequency meter.**

Frequency Meter - Indicates the output frequency in Hertz of the generator selected by the **frequency switch.**

Synchronizing Switch - Selects the generator whose frequency to be monitored by the **synchroscope.** This generator is usually the one being brought on line.

Synchroscope - Indicates whether the output frequency of the generator selected by the **synchronizing switch** is faster or slower than the frequency of the bus. Counterclockwise rotation of the needle means that the selected generator's output frequency is slower than the frequency of the bus. Clockwise rotation of the needle means that the selected generator's output frequency is faster than the frequency of the bus. A clockwise rotation is recommended for synchronizing.

Synchronizing Lights - Indicates the phase relationship between the generator selected by the **synchronizing switch** and the generator(s) already on line. The relative brightness of the lights indicates the degree that the generators are out of phase. When both lights are brightly lighted, the two generators are 180 degrees out of phase. When both lights are out, the two generators are in phase.

Breaker Close Button - When pressed, closes the circuit breaker of the generator selected by the **synchronizing switch** and brings the generator output on line to the 600 VAC bus. The generator's circuit breaker must be in the CHARGED position before the **breaker close pushbutton** will work.

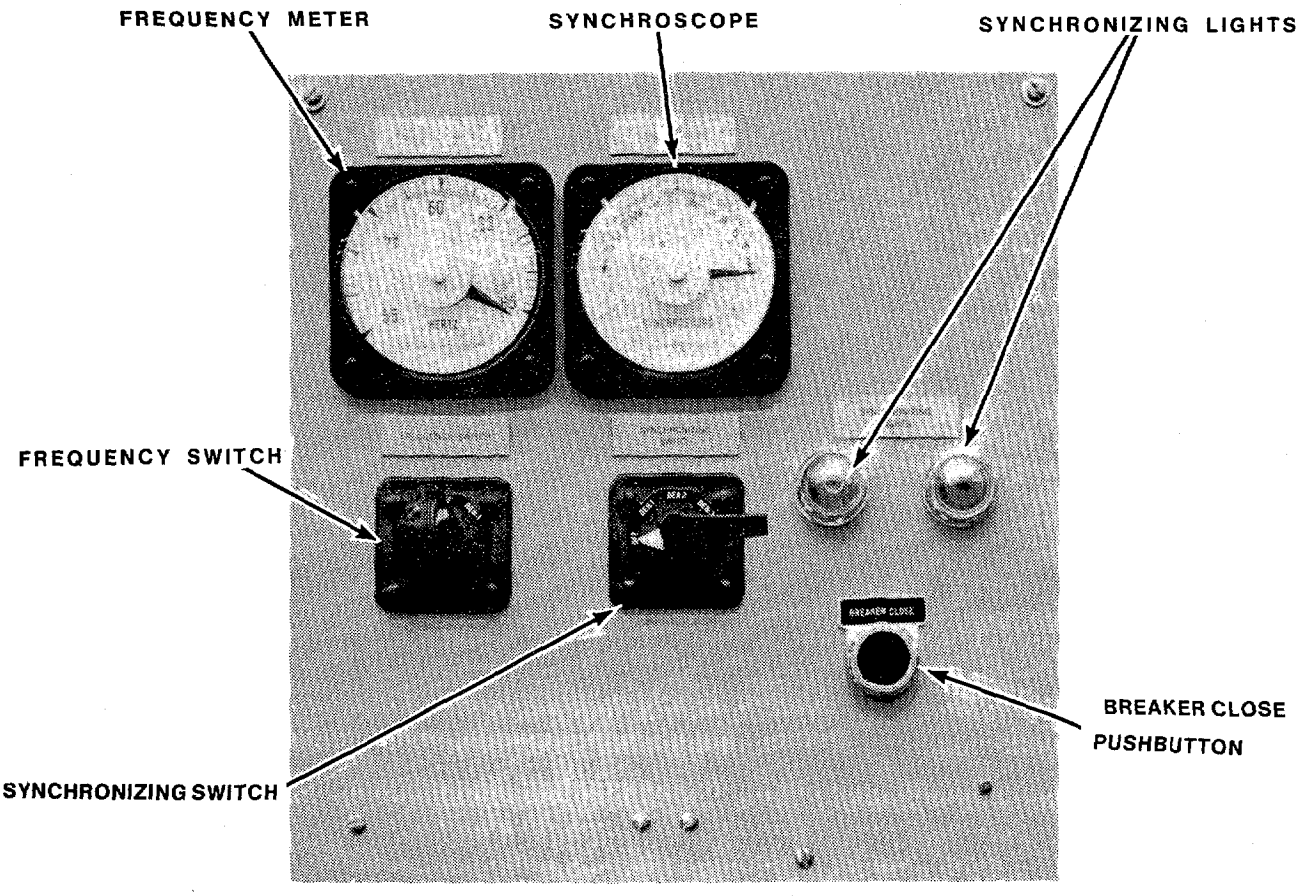


Figure 7.
Synchronizing Panel Controls and Indicators

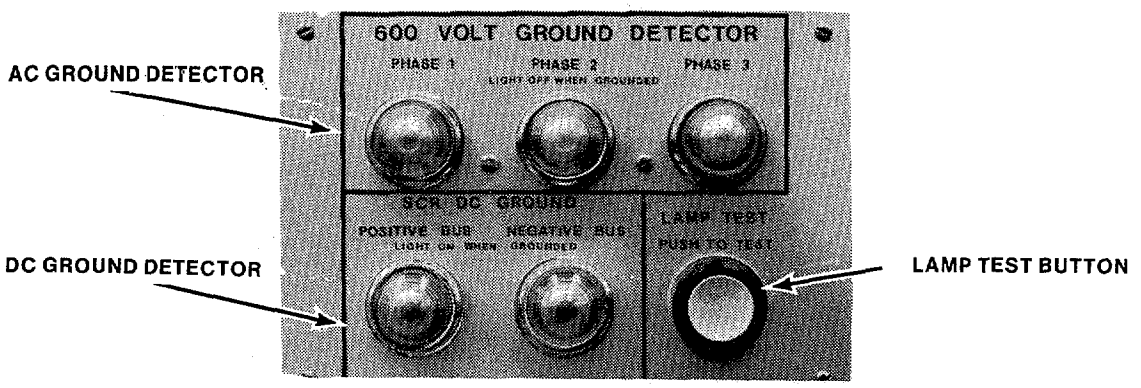
GROUND DETECTOR MODULE

This module, shown in Figure 8, detects ground faults within the AC and DC power system and displays the faults on the lights described below. A lamp test button allows the operator to check the lamps.

AC Ground Detector - These three lights correspond to the three phases present on the 600 volt AC bus. The lights are normally on; when a phase is grounded, the corresponding light turns off.

DC Ground Detector - These two lights correspond to the positive and negative SCR DC buses. The lights are normally off; when a bus is grounded, the corresponding light comes on.

Lamp Test Button - Depressing this button causes all ground detector lamps to light. This should be done periodically to detect faulty lamps.



(DSF53915)

Figure 8.
Ground Detector Module

FEEDERS

The **motor control center circuit breaker** connects or disconnects power to the **motor control center**, which is a collection of starters for auxiliary rig motors such as blowers and lube motors. The **lighting transformer circuit breaker** connects or disconnects power to the **lighting transformer**, which supplies power for lights and convenience outlets on the rig.

BATTERY MODULE

This module is located in the **contactor panel**. It is used to energize the controls of the first generator module to be started up. The **battery module power switch** should always be in the ON position during rig operation so that the battery can be recharged. The switch must be turned OFF when the rig is shut down to be moved.

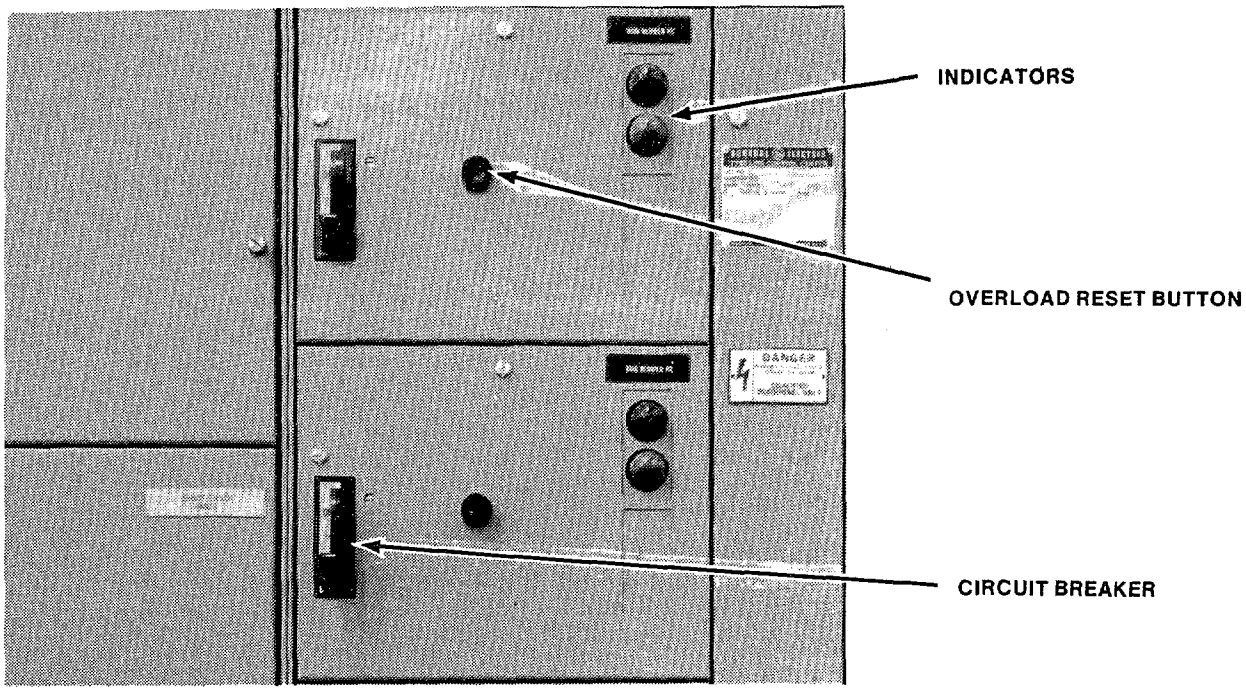
MOTOR CONTROL CENTER

The Motor Control Center (if supplied with the lineup) contains starters for auxiliary AC motors used on the rig. These auxiliary motors would typically run blowers, supercharger pumps, and mud pump oilers. The Motor Control Center Circuit Breaker provides 600 VAC from the main generator bus to the Motor Control Center. If the motor starters require 480 VAC, a transformer steps down the generator voltage from 600 VAC to 480 VAC. Figure 9 is a photograph of a portion of a typical Motor Control Center showing the following controls and indicators:

Circuit Breaker - This is a three position switch that connects or disconnects power to the starter. In the ON position, power is applied to the starter. If the circuit breaker is in the TRIP position, AC power has been removed from the starter because of an overcurrent condition. To reset the circuit breaker, move the switch to the OFF position and then back to the ON position.

Overload Reset Button - This button resets a starter that opened because of an overload. Since an overload will not trip the circuit breaker, if the motor stops and the circuit breaker has not tripped, wait five minutes and then press the overload reset button to reset the starter.

Indicators - The red and green (optional) indicators show whether or not the breaker is closed and the motor is running. If the green light is on, the breaker is closed, but the motor is not running. If the red light is on, the motor is running. If neither light is on, the breaker is open.



(DSF54328)

Figure 9.
Portion of Typical Motor Control Center

POWER CONVERSION PANEL

The power conversion panel uses SCRs (Silicon Controlled Rectifiers) to convert three phase, 600 VAC supplied by the generators to a variable DC voltage that can be used by the drilling motors. As shown in Figure 10, the front of this panel contains two indicators and two meters that monitor the operation of the SCRs:

Voltmeter - This meter indicates the DC voltage supplied to the assigned motor by the SCR.

Ammeter - This meter indicates the DC current supplied to the assigned motor by the SCR.

Suppression Light - When ON, indicates that the SCR shut down because of an instantaneous overcurrent condition.

Blower Loss Light - When ON, indicates that the SCR has been shut down because the SCR blower is not operating. If the motor is still running, this light indicates that the SCR temperature is too high.

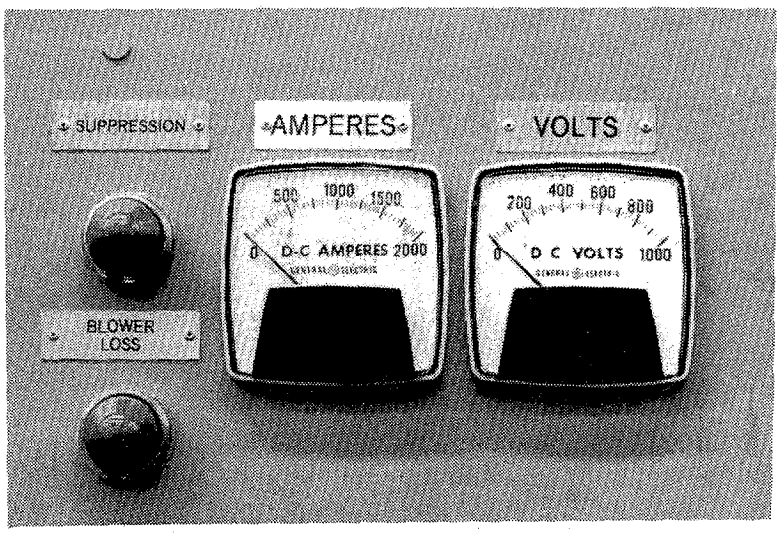


Figure 10.
Power Conversion Panel Indicators

CHAPTER 3
OPERATING PROCEDURES

The procedures outlined below are for normal, day-to-day operation of the μdrill™ 1500 rig drive system, including rig start up, operation, and shut down.

These procedures assume a rig drive system with typical options. Your system may not have some of the controls or indicators mentioned. In addition, rig drive systems vary in the number of generators, generator controls, power control panels, and motors. Disregard any steps that do not apply to your rig drive system.

In each of the steps below, the names of modules, panels, indicators, and controls which are described in this manual are printed in **bold type**. For descriptions of other devices, refer to the appropriate vendor publications.

RIG START UP

This procedure describes how to start up a typical μdrill™ 1500 rig drive system. It assumes that the rig, drive system, generators, and engines have been properly installed and checked out.

1. Preliminary steps:
    a. Ensure that the **plant shutdown** and **emergency stop** switches on the **driller's console** are pulled out.
    b. Remove all assignments at the **driller's console** and turn throttles off.
    c. Position the **battery module power switch** to the ON position.
    d. Ensure that the **engine control switch** for the generator being started is in the IDLE position.

2. Start the desired engine and Follow the engine warm up procedure before continuing with the following steps. (The following steps assume that generator 1 is being brought on line first, followed by generator 2.)

3. Make sure the circuit breaker behind the **synchronizing panel** door is ON. Set the **synchronizing switch** on the **synchronizing panel** to GEN 1.

4. Set the generator 1 **engine control switch** to RUN. All of the indicating lights on the generator 1 **AC generator control panel** should come ON (except Gen ON light). Check that the **generator voltmeter** for generator 1 reads 600 volts.

5. Set the **frequency switch** on the **synchronization panel** to GEN 1. Turn the **frequency trim knob** until the **frequency meter** reads 60 Hertz.

6. Set the generator 1 **engine control switch** to LOAD.

7. Charge the **generator breaker** for generator 1 and depress the **breaker close switch** to bring the generator on line to the AC bus.

8. Close the **lighting transformer circuit breaker** and the **motor control center circuit breaker**. Start the essential motors from the **motor control center** (i.e., air compressors, fuel pumps, radiator fans, etc.).

9. Start the generator 2 engine (as required) according to the instructions in steps 2 thru 6 above. Check the **generator voltmeter** for generator 2 and make sure it reads 600 volts.

10. You will now use the **synchroscope** to synchronize the output frequency of the two generators before you bring Generator 2 on line. The **synchroscope** indicates whether the output frequency of the generator selected by the **synchronizing switch** is higher or lower than the output frequency of the generator already on line. Set the **synchronizing switch** on the **synchronizing panel** to GEN 2. Adjust the **frequency trim knob**, if necessary, on the generator 2 **AC generator control panel** until the **synchroscope** needle is rotating slowly (about one revolution every 15 seconds).

11. Charge the **main generator breaker** for Generator 2.

12. Watch the **synchroscope** and depress the **breaker close switch** just before the rotating needle reaches the twelve o'clock position. This brings Generator 2 on line to the AC bus. Synchronizing lights should be turning off.

13. The meters on the generator 2 **AC generator control panel** will now indicate Generator 2 voltage, current, and power. The values on these meters should be equal for both generators.

14. If your rig uses three generators, repeat steps 9 through 13 for the third generator

15. Ensure that all **throttles** on both the **driller's console** and **mud pump console** are OFF.

15. Check that the **AC disconnect switch** on the front of each **power control panel** are closed to provide AC power to the SCRs and control logic. At this point the **suppression light** and **blower loss light** should be OFF and the SCRs are ready to have motors assigned to them.

16. Energize the appropriate auxiliary motor starters from the **motor control center**. These would include drill motor blowers, mud pump oiler motors, and so forth. These motors will start automatically when the drilling motor assignments are made from the **driller's console**.

17. Check that motor cutout switch is not pinned.

The rig drive system is now ready for operation.

RIG OPERATION

Once the rig drive system has been started, the drilling motors are controlled from the driller's console and mud pump console. The operator will seldom need to concern himself with the control equipment.

Most operations can be broken down into four types of activities:

- Making motor assignments
- Controlling motor speed and direction
- Monitoring rig operation
- Correcting operating problems

These activities are described below. As we describe each activity, refer back to the photographs of the driller's console (Figure 2) and mud pump console (Figure 3).

MAKING MOTOR ASSIGNMENTS

The **assignment switches** are used to select the motor that will be assigned (connected) to each SCR power converter. Various combinations of motors and SCRs are possible. Each **assignment switch** represents one SCR power converter and the switch position determines which motor will be connected to that SCR.

Before an assignment can be made, be sure to check the following items:

1. The **SCR On** light for the desired SCR power converter must be ON.
2. The drilling motor to be assigned must not already be assigned to another SCR.
3. The drilling motor auxiliary switch (located on the motor) must not be depressed.
4. The **throttle** corresponding to the **assignment switch** must be in the OFF position.

Once the motor has been assigned, the **Blwr On** light for that motor will come on and the motor can be controlled via the **throttle**.

CONTROLLING MOTOR SPEED AND DIRECTION

The drawworks and rotary table are usually driven by the same motor, which is powered by one SCR power converter. The drawworks speed is determined by the appropriate **throttle** on the **driller's console**, although a **foot throttle** may also be used for this purpose. The motor runs at the highest speed setting of the two **throttles**. A usual practice is to set cathead speed with the hand **throttle** and use the **foot throttle** to control the hoist speed. The **rotary table torque meter** indicates the torque applied to the drill pipe by the rotary table, while the **rotary table torque limit control** limits the torque produced by the drawworks/rotary table motor any time the rotary table clutch is engaged. This torque limit should be set according to the pipe grade and diameter to prevent twisting off the drill pipe during operation.

The drawworks/rotary table motor direction is controlled by the **drawworks direction switch**, but the direction can be changed only when the drawworks **throttle** is in the OFF position. If the **drawworks direction switch** is placed in the FWD REG position (optional), regenerative braking occurs after the **foot throttle** is released. Regenerative braking is a faster method of slowing down a motor by pumping its energy back into the AC power system. It is used to improve tripping operations by allowing the cathead to be engaged sooner.

Most rigs have two mud pumps, each powered by a separate SCR power converter. If the rig has a **mud pump console**, the mud pumps can be controlled from either the **driller's console** or the **mud pump console**. Setting the **mud pump control here - remote switch** to the REMOTE position gives control of the mud pumps to the **mud pump console**; at the same time, the **MP control here** lights on the **mud pump console** will light. Before operating the mud pumps, the **MP blower** and **MP aux** lights should be ON, indicating that the blowers and lube motors are operating. When the mud pump **throttles** are moved from the OFF position, there may be a short delay before the mud pumps start. This delay will occur only if the rig uses a supercharger motor in conjunction with the mud pumps; it takes a short time for the supercharger motor to be started automatically by the system.

MONITORING RIG OPERATION

There are several indicators and meters on the consoles that should be monitored during rig operation. Make a habit of occasionally scanning the console indicators and meters to ensure proper rig operation and early detection of problems.

Should any of the **status lights** show an abnormal condition the **audible alarm** will sound, unless the **alarm silence switch** is ON. Leave the silence switch OFF unless you need to silence an active alarm while you fix a problem.

Before starting a drilling operation or making a function change, the driller should check the following things:

1. Are the correct number of **gen on** and **SCR on** lights ON?
2. Are all **throttles** in the OFF position?
3. Is the **rotary table torque limit control** set according to the drill pipe grade and diameter?
4. Are the **assignment switches** and **direction switch** set properly?

CORRECTING OPERATING PROBLEMS

Several common operating problems are listed below along with suggestions for correcting them:

Insufficient Rig Power - If the speed of all DC motors decreases and the **KVA/KW limit** light comes on, there are not enough engines/generators on line to support the heavy drilling load. Should this occur, the load must be reduced or an additional generator brought on line.

Insufficient RT Torque - If only the rotary table's speed decreases, the torque limit set by the **rotary table torque limit control** has been reached. Ensure that the limit control is set properly for the type of pipe being used. If the setting is correct, you must reduce the torque requirements. There may be too much weight on the bit or the wrong gear may have been selected.

Unable to Assign Motor - When a motor has been successfully assigned to an SCR, the appropriate **blwr on** light should come on after a short delay. If this does not occur, the assignment has failed. Check that the appropriate **throttle** is OFF, the proper **SCR on** light is on, and that the motor's blower is operating. If necessary, try to assign the motor to a different SCR. If the motor still cannot be assigned, check the **motor control center** starters for that motor, along with the motor auxiliary switch, motor pressure switch and field exciter breakers (for shunt motors).

RIG SHUT DOWN

This procedure describes how to completely shut down the μdrill™ 1500 rig drive system in preparation for power off maintenance or moving of the rig.

1. Turn all **throttles** on the **driller's console** to the OFF position. Turn all **assignment switches** to the OFF position.

2. Stop all non-essential loads.

3. Turn the **engine control switch** on the **AC generator control panel** to the RUN position. This unloads the engine. Turn the **generator circuit breaker** to the OFF position. Repeat this step for each generator.

4. Turn the the **engine control switch** on each **AC generator control panel** to the IDLE position. After the proper idle time, turn the **engine control switch** to the OFF position.

5. Turn the **battery module power switch** to the OFF position.

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